

Systematic Literature Review

Introduction

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Introduction

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What is Systematic Literature Review?

Systematic Literature Review:

Is a review of a clearly formulated question(s) that uses systematic and explicit methods to **identify, select and critically appraise relevant research** and to **collect and analyse** data from the studies that are included in the review.

(Cochrane Collaboration, 2014).

Why Systematic Literature Review?

SLR aims to address the problem being researched by **identifying**, **critically evaluating** and **integrating** the findings of all relevant **high-quality individual studies**.

Systematic Literature Review?

A good SR should achieve the following;

1. Establish to what extent existing research has progressed towards clarifying a problem,
2. Identify relations, contradictions, gaps and inconsistencies in the literature, and explore reasons.
3. Comment on, evaluate, extend, or develop theory.
4. Provide implications for practice and policy.
5. Describe directions for future research.

(Baumeister & Leary, 1997; Bem, 1995; Cooper, 2003)

Systematic Literature Review?

A good SLR is characterised by being;

- **Objective:** *Explains criteria for inclusion & exclusion of findings.*
- **Systematic:** *Methodical & organised.*
- **Transparent:** *Clear judgement on inclusion & exclusion.*
- **Replicable:** *Allows for update and further review at a later stage.*

SLR should adopt a JUDGE AND JURY approach, rather than JUDGE AND LAWYER approach.

How do I do a SLR?

Key stages in conducting a SLR.

1. Scoping,
2. Planning,
3. Identification/ searching,
4. Screening,
5. Eligibility.

Scoping

Formulate your research questions

Example:

1. What can be done to improve the currently used workload framework?
2. How best can we implement a real-time system to detect and prevent fraud?

This should help you to be focused and narrow your literature search.

Planning

Break your research questions down into individual concepts to create search terms.

- Identify key terms/ ideas from the RQ, *(eg: workload frameworks)*

Formulate preliminary inclusion & exclusion criteria

- Based on your RQ *(eg: only real-time systems are considered)*

Make a record of the details of your searches and results obtained.

Identification /Searching

- Select databases that are relevant to your topic.
- Consider using Boolean search operators (NOT, AND, OR).
- Consider searching by publication year.
- Consider additional searches on alternative sources to ensure diversity.
- Consider published works.

Screening

- Read titles and abstracts of identified articles.
- Export references of selected articles to a citation manager (EndNote, RefWorks, Ms Word, etc.)

Eligibility

- Sift the full-text version of potentially eligible articles and extract relevant information.
- Add the included material onto your rough SLR document.
- Paraphrase and reference.
- Use turnitin to check your plagiarism level.

How to Present a Systematic Review

- Provide a theoretical and empirical background to the literature, explaining key terms, definitions and concepts;
- Describe the characteristics of included studies.
- Critically evaluate and integrate the results in an **unbiased and systematic way**, including stating whether the results are heterogeneous and discussing possible reasons.
- Balance conflict resolution:

(by identifying inconsistencies in study results & points of contention in theories, conceptualizations and methods in the literature)

Cont...

- Summarize and discuss the findings and conclusions of the review in a balanced and impartial way.
- Explicitly and intuitively link your conclusions to the evidence reviewed.
- Discuss the strengths and limitations of the literature articles.
- Establish to what extent existing research has progressed towards clarifying a particular problem.
- Comment on, evaluate, extend, or propose a new theory.
- Draw conclusions and make recommendations for practice

Common Errors with literature reviews

Common Errors

- **List-like writing that lacks synthesis**
- Relying on material that is likely to be out-of-date
- Exclusion of landmark studies
- Lack of a clear organisational structure
- Not discriminating between relevant and irrelevant materials
- **Not being sufficiently critical**

List-like Writing that Lacks Synthesis

Resulting from failing to integrate arguments into the review.



DO NOT SUMMARISE YOUR READINGS.

Example 1:

During the past decade, many researchers have shown interest in computer security. Their findings are reported here. Smith (2004) developed a model investigation on *(the next paragraph describes Smith's model)*... Jones (2005) *sees that...* *(the following paragraph says what Jones asserted)*... Harry (2008) makes a clear distinction between... *(the following paragraph states what Harry says)*.

Example 2:

This study brings together work in two areas of computer security. According to Robbins et al (1992) the early theories of computer security were concerned with unauthorised access.

Current thinking however does not discount these theories, but simply builds upon them to include contemporary approaches. Scholl (2007), argue that there are three basic approaches to security. He proposed that these approaches can be seen as an elaboration of Peter (2014)'s original ideas...

Not being sufficiently critical

Note that, the purpose of a literature review:

1. Is not just to summarise what is currently known about a topic.
2. Is to provide a detailed justification for your research.

That is, ultimately it is in the form of an argument and this argument is developed by pointing out "holes" or "Gaps" in the reviewed sources.

The End.

Thank you for attending, hope you
learnt something.